2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), 13-15 February 2025, CUET, Chattogram-4349, Bangladesh

2025 International Conference on Electrical, Computer and Communication Engineering (ECCE) | 979-8-3503-5750-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ECCE64574.2025.11013347

Improvement of Fault Finding Techniques in Industrial Robots: Using Unsupervised Learning for Predictive Maintenance

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***Abstract*—A detailed comparative analysis of various machine learning models for predictive maintenance in industrial robotic systems is conducted. The focus is on constructing accurate models to forecast industrial robot breakdowns, enabling timely repairs, and reducing downtime. Data from industrial robots is gathered, feature engineering is performed, and multiple machine learning models are analyzed using the Silhouette Score. The goal is to identify the most effective model for predictive maintenance, aligning with the need for optimal system performance.**

Keywords—Predictive Maintenance, Machine Learning, Industrial Robots, Silhouette Score, Feature Engineering.

I. Introduction

A. Motivation

The deployment of robots in industrial settings has become increasingly significant due to rising manufacturing costs, job complexity, and hazards. The International Federation of Robotics has indicated significant growth in robot usage across various industrial sectors, including semiconductor and automobile manufacturing. The primary benefits of robotics include enhanced operational efficiency, improved product quality, and cost savings. However, the increased use of industrial robots also raises the potential for unexpected malfunctions, leading to significant downtime, financial losses, and worker casualties. Effective maintenance strategies are crucial to avoid these disruptions [1].

Current maintenance practices for industrial robots typically involve schedule-driven maintenance, which can cause unnecessary downtime, or reactive fail-and-fix maintenance, which may result in severe unplanned downtime. Predictive maintenance (PdM) aims to anticipate equipment failures and schedule maintenance proactively, minimizing operational losses. Machine learning (ML) techniques have proven effective for PdM, leveraging historical performance data and sensor readings to predict failures [2].

B. History and Background

1) Maintenance

Maintenance ensures the proper and consistent functioning of machines, robots, systems, and equipment through various strategies aimed at avoiding failures, reducing downtime, extending asset life, and improving efficiency. The primary forms of maintenance include:

- **Reactive/Corrective Maintenance**: Repair or replacement of defective equipment after failure.
- **Preventive/Planned Maintenance (PM)**: Scheduled operations to prevent equipment failures.
- **Predictive Maintenance (PdM)**: Uses data analysis to predict equipment breakdowns.
- **Pro-Active Maintenance**: Real-time monitoring and servicing based on specific conditions.

Fig. 1. Types of Maintenance [3].

2) Machine Learning

Machine learning (ML), a subset of AI, is developing statistical models and algorithms that enable computers to learn from data and make predictions or judgments without explicit programming. ML techniques are divided into:

- **Supervised Learning**: Uses labeled data to predict or classify.
- **Unsupervised Learning**: Identifies patterns and structures in unlabeled data.
- **Reinforcement Learning**: involves achieving goals through interaction with the environment and feedback.

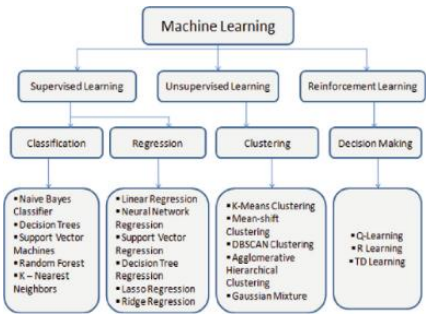

Fig. 2. Machine Learning Techniques Categorizations [4].

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C. *Objectives*

- Determine the best method to forecast the condition of industrial robots.
- Construct a reliable prediction model for precise and timely maintenance planning.
- Minimize robot downtime to ensure continuous and effective output.

D. *Scopes*

This work focuses on fault detection in industrial settings, with potential applications in robotics, manufacturing equipment, automation, and management structures. The goal is to develop robust algorithms for early detection of anomalies, enhancing safety, reliability, and productivity.

E. *Conclusion*

This chapter provides an overview of the motivation, historical background, maintenance strategies, and machine learning techniques relevant to the study. It outlines the thesis objectives and scope, emphasizing the importance of predictive maintenance in industrial robotics.

II. Literature Review

Several studies have explored different methods for predictive maintenance using machine learning in industrial robot.

The study "Unsupervised Fault Detection of Industrial Robot Joints Using Current Signal" by Fu et al. [5] focuses a method for detecting faults without supervision using spectral clustering and current signal sensitivity. The method divides collected signals into parts and uses time-domain features based on sensitivity to locate faults. The procedure is validated with an industrial robot that has been tested for reliability.

The study "Data-driven gearbox failure detection in industrial robots" by Orkisz et al. [2] analyzes the impact of four preprocessing techniques on detecting gearbox breakdowns on industrial robots. Results show that these techniques improve classification performance by varying variance in AUC across all methods, including local variation and estimated measures. The LR method's performance also improves significantly with PCA preprocessing. However, these techniques may face challenges in assessing new robots due to repeated processes result in greater measurement variation. The study aims to address these issues by increasing task invariance, emphasizing temporal fluctuations, and isolating key elements.

The study "GAN-based Data Augmentation Strategy for Sensor Anomaly Detection in Industrial Robots" by Lu et al. [6] focuses on a new approach of detecting anomalies framework for industrial robots, with an offline training step for model creation and an online interpretation stage for immediate anomaly detection. The authors employ incremental learning to increase detection accuracy for a new form of anomalous in the online judgment state. They introduce an enhanced GAN algorithm called MSGAN using WGAN-GP to enhance anomalous samples and speed up training convergence. A genuine dataset of robotic sensors was used to validate the framework and the MSGAN algorithm, showing effectiveness and improved detection accuracy.

The study "A Hybrid Approach for Improving Unsupervised Fault Detection for Robotic Systems" by Khalastchi et al. [7] proposes a hybrid approach, where an unsupervised approach labels a low-inaccuracy data set, and a supervised approach uses the labeled data offline to create an online FD model. The paper aims to validate the choice between unsupervised and hybrid fault detectors, as there is no priori labeled data. The work's primary contributions include a theoretical analysis of the circumstances in which the combination method is anticipated to be more accurate, supported by empirical analysis utilizing data sets from three distinct robotic domains: a commercial Unmanned Aerial Vehicle (UAV), a laboratory robot, and a high loyalty flight simulator. The hybrid technique improves different unsupervised FD approaches and fits well with a prediction tool.

The study "High-Accuracy Unsupervised Fault Detection of Industrial Robots Using Current Signal Analysis" by Cheng et al. [1] presents a novel unsupervised defect detection system for the industrial robots, utilizing a physics-based framework and GMM. A signal analysis module to gather analyzed signals and differentiate robot motion phases from recorded current signals. Removal of Features Parts extract robust fault patterns that are responsive to defects but not to robot movement. In both steady and transient states, STFT and HT extract two motion-insensitive fault characteristics, SNR and MFI. Unsupervised learning methods are utilized to the extracted fault detection features.

The study "Fault detection and state estimation in robotic automatic control using machine learning" by Natarajan et al. [8] presents an approach to autonomous control networks that uses leakage and state estimation using the Gaussian filter (GF) and the Accelerated Gradient Descent-based vector machine (AGDSVM). The method addresses the limitations of traditional sliding mode controllers, such as asymptotic convergence and chattering. The study also proposes a fault detection and exclusion technique using BD, or Bhattacharya Distance, is used to estimate states. The research also explores the use of numbers to distinguish encoder data and extended Kalman Filter for kinematic state estimators. The research also proposes Multiple-order-holder (MOH) for fault detection and parameterizes the "occurrence frequency" using interval durations.

The work "Fault diagnosis of industrial robot based on dual-module attention convolutional neural network" by Lu et al. [9] a DMA-CNN model is proposed for defect diagnostics in industrial robots, identifying fault characteristics through various attention mechanisms and improving convolutional neural network extraction capacity. The model uses a sample dataset, constructs a DMA-CNN for feature extraction, and tests on genuine data from industrial robots. The model successfully detects abnormal state and outperforms Other models differ in accuracy, F1_score, precession, and recall, introducing a novel and efficient method for fault diagnostics.

The work "Unsupervised Fault Detection in Varying Operating Conditions" by Michau and Fink [10] compares five unsupervised feature alignment networks (UFAN) for Fault detection in complicated industrial systems. The baseline approach, based on two months of data, is the least performing and has limitations. Three ways rely on matching the unit of concern with another healthier unit to improve learning of the single class classifier. The UFAN method is the top performing, as it becomes valid models for more than half of all potential pairings.

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The study "CASPER: Context-Aware Anomaly Detection System for Industrial Robotic Arms" by Moran introduces CASPER, a context-aware machine learning-based anomaly detection infrastructure that monitors an industrial robotic arm's movements, reaching a precise level of 97% and F1 score of 97% [11].

The paper " Mechanical fault detection based on machine learning for robotic RV reducer using electrical current signature analysis: a data-driven approach" by Raouf et al. [12] introduces a complete approach to robots RV reducer malfunction detection using Machine Learning (ML)-based classification. The method uses electric current data from different rotation speeds with Healthy, Faulty, and Faulty Aging reducers. Various features extracted, and correlation analysis and the chi-square test are applied to reduce redundant information. The chi-square test shows significant results compared to correlation analysis. Classifiers like NB, KNN, SVM, and ensemble are used.

The study "A Research Study on Unsupervised Machine Learning Algorithms for Early Fault Detection in Predictive Maintenance" by Amruthnath and Gupta. The research compared machine learning techniques for unsupervised learning-based early fault detection. T^2 Statistic yielded more precise results than GMM method, allowing for fault identification even in manufacturing environments with minimal domain knowledge. Clustering methodology is better for detecting different levels of faults, especially when machine maintenance is expensive. T^2 statistic is an excellent tool for fault detection, but clustering algorithms are better for different levels of fault detection [13].

The study "Machine Learning approach for Predictive Maintenance in Industry 4.0" by Paolanti et al. [14] focuses on a cutting machine with a revolutionary PdM approach based on learning through machines, aimed at minimizing downtime and costs. The methodology uses data collected from sensors, PLCs, and communication protocols to create dynamic decision rules for maintenance management. Preliminary results show high accuracy (95%) in predicting machine states on a 530731 data set of 15 machine features.

The study titled "Adoption of machine learning technology for failure prediction in industrial maintenance: A systematic review" by Leukel et al. [15] explores the use of machine learning (ML) technology for failure prediction technique in industrial maintenance. It identifies barriers to knowledge accumulation and suggests future research to improve technology for predicting failure in industrial maintenance. The review also suggests new practices to interpret and compare results from single studies, thereby enhancing the understanding of how ML technology can benefit failure prediction.

In the work "Selecting an Appropriate Supervised Machine Learning Algorithm for Predictive Maintenance" by Ouadah et al. [16] the purpose was to increase predictive maintenance by Choosing the most suitable supervised machine learning algorithm. They examined Random Forest, Decision Tree, and KNN using real-world and simulation data, finding similar performance for Random Forest and Decision Tree, but KNN excelled with comprehensive data.

The study "Machine Learning Algorithms for Smart Data Analysis in Internet of Things Environment: Taxonomies and Research Trends" by Alsharif et al. [17] stressed the usefulness of machine learning techniques for IoT symmetric applications. It focused on supervised and unsupervised ML, covering their advantages, limitations, sand research trends for smart data analysis in IoT. Recommendations for additional study were made.

The study "A predictive maintenance model for health assessment of an assembly robot based on machine learning in the context of smart plant " by Chakroun et al. [18] introduces a machine learning-based predictive maintenance approach that employs a Bayesian filter to evaluate and anticipate the decline of power transmission in assembly robots. The model is designed to assure the operational preparedness of manufacturing units and improve production efficiency. The model aligns with Industry 4.0 technologies and uses real-time data-collecting sensors in assembly units. The data undergoes in-depth analysis, including expert knowledge acquisition, descriptive analysis, bivariate analysis, and study of interactions with the configuration factors. The model, DBF, is used to estimate the transmitters' degradation state and predict their future conditions.

The study "Development of Predictive Maintenance Models for a Packaging Robot Based on Machine Learning" by Chakroun et al. [19] presents a predictive model for power transmitter machine learning techniques are used to assess health in a packing robot. Using Discrete Bayesian Filter (DBF) and Naive Bayes Filter (NBF), it outperforms the NBF model in predictive power, enabling better maintenance scheduling and cost reductions in Industry 4.0.

This section discussed major contributions to the subject of predictive maintenance for industrial robots, focusing on the efficacy of machine learning techniques in fault identification and predictive modeling.

III. Methodology

A. Introduction

This chapter outlines the methodology used to conduct the comparative analysis of machine learning models for predictive maintenance. The process includes data collection, preprocessing, model development, and evaluation.

B. Different Stages of Methodology

1) Data Collection and Preprocessing

Data were collected from industrial robotic arm(UR3e). The preprocessing steps included handling missing data, data standardization, and feature extraction.

Fig. 3. Industrial Robotic Arm(UR3e). Data were collected from this robot by change the arm's motion by modifying the joint velocity to create anomalies.[11]

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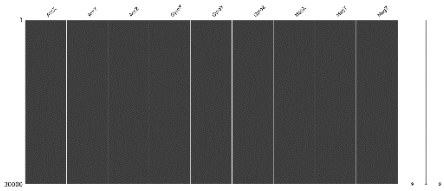

Fig. 4. No missing data.

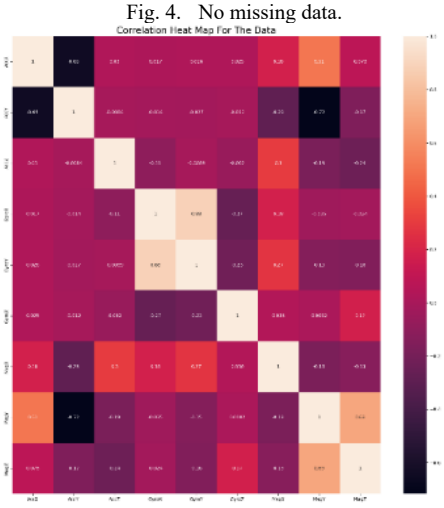

Fig. 5. Correlation Heatmap of the Used Data.

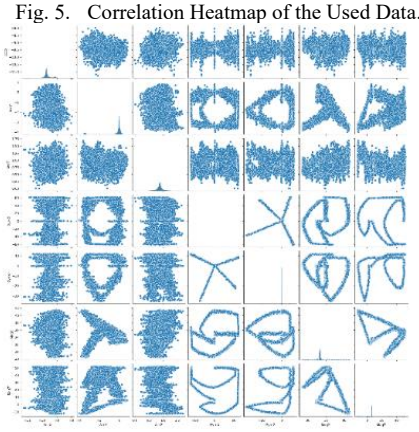

Fig. 6. Pairplot Visualization.

2) *Model Development*

Various machine learning models were developed and trained using the collected data. The models were evaluated based on their performance in predicting robot breakdowns.

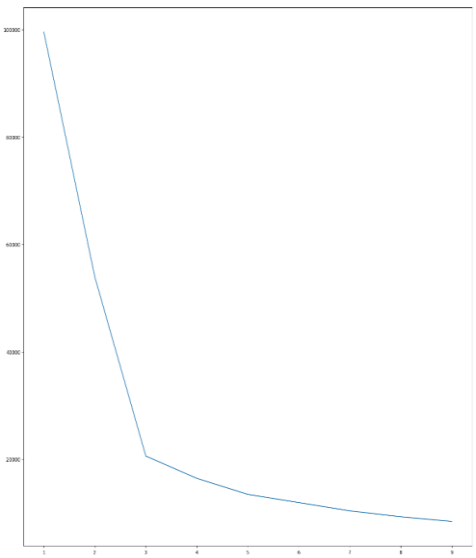

Fig. 7. Elbow Curve. Elbow Curve shows about 3 cluster was best for this model.

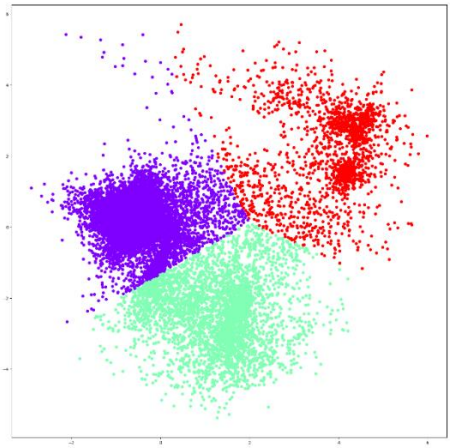

Fig. 8. K-Means Cluster. Figure shows Purple is normal state, Green is anomalous state, Red is outliers(anomalous state with higher joint velocities and opposite).

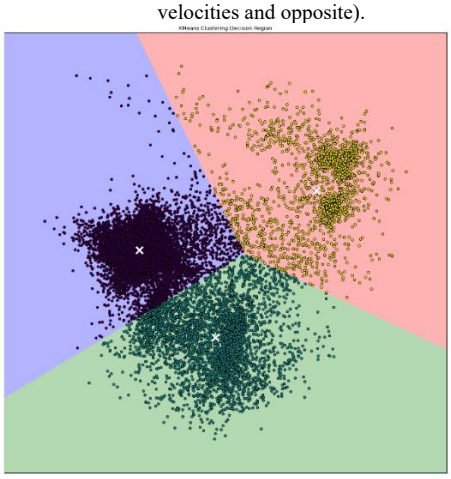

Fig. 9. K-Means Clustering Decision Region. The decision region implies whether the upcoming value is normal or anomalous.

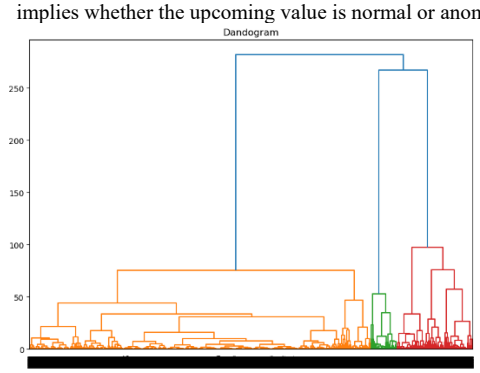

Fig. 10. Dendrogram

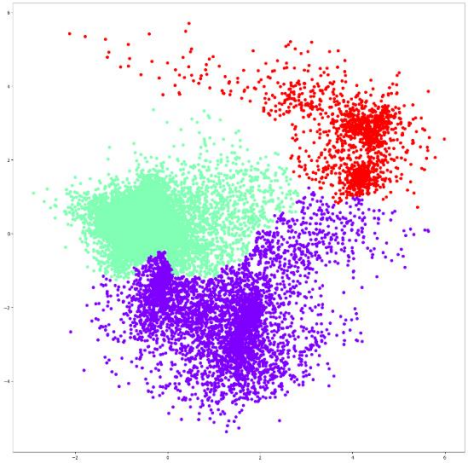

Fig. 11. Agglomerative Cluster

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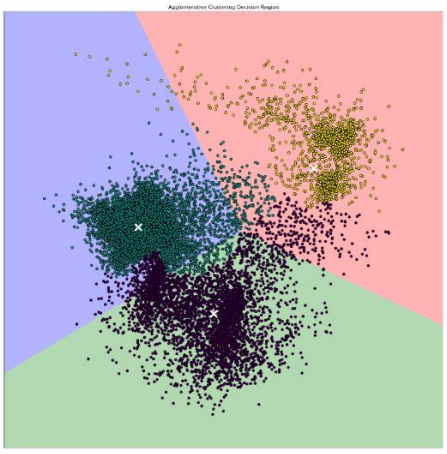

Fig. 12. Agglomerative Clustering Decision Region

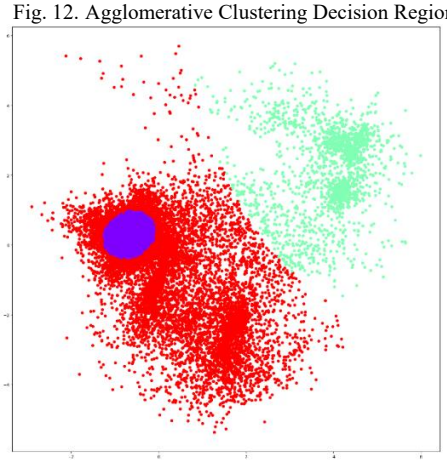

Fig. 13. Gaussian Mixture

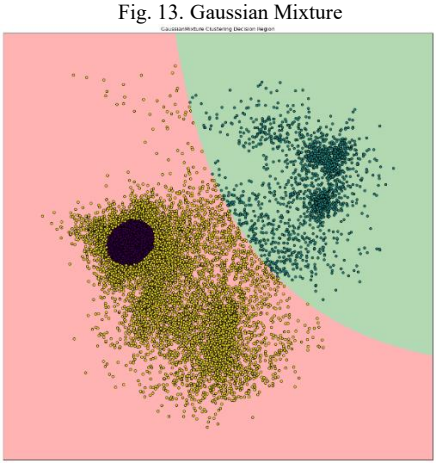

Fig. 14. Gaussian Mixture Clustering Decision Region

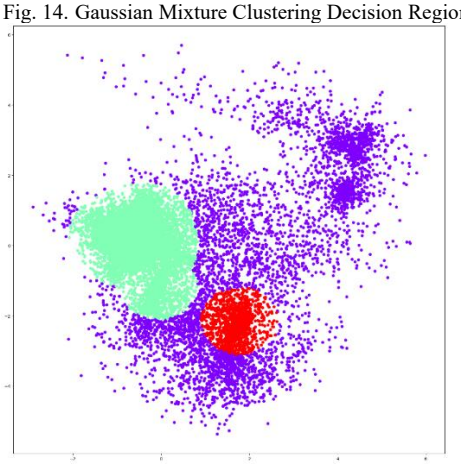

Fig. 15. DBSCAN

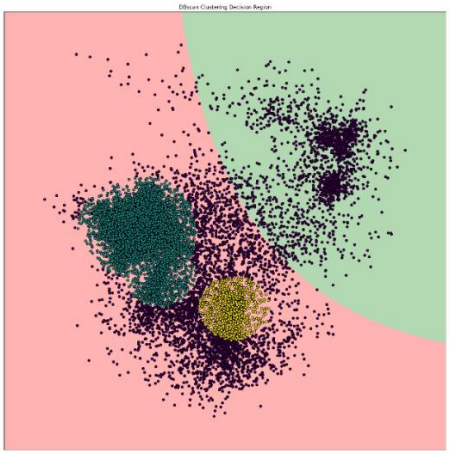

Fig. 16. DBSCAN Clustering Decision Region

IV. Results and Discussion

A. Result Evaluation

The performance of different machine learning models was evaluated using the Silhouette Score, a metric for assessing the quality of clustering.

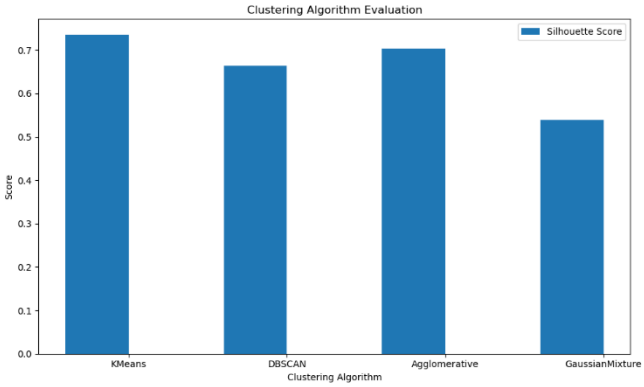

Fig. 17. Silhouette Score

B. Silhouette Score Comparison

TABLE 1. Silhouette Score

Model	Silhouette Score
1. K-Means Cluster	0.74 (Approximately)
2. DBSCAN	0.67 (Approximately)
3. Agglomerative Cluster	0.70 (Approximately)
4. Gaussian Mixture	0.54 (Approximately)

A higher silhouette score indicates better clustering performance, which was the primary parameter used to evaluate the efficacy of clustering algorithms for predictive maintenance of industrial robotic systems. Table 1 demonstrates that the K-means approach has the highest clustering quality, with a silhouette score of 0.74 (approximately). The Gaussian Mixture approach, on the other hand, yielded the lowest silhouette score of 0.54 (approximately), suggesting less exact categorization. As a result, the DBSCAN approach was the best-fitting model based only on the silhouette score from Table 1.

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C. Discussion

The analysis revealed that certain machine learning models outperformed others in predicting industrial robot breakdowns. The discussion focuses on the implications of these findings for predictive maintenance.

V. Conclusion

This study contributes to the field of predictive maintenance by providing a comparative analysis of machine learning models for fault detection in industrial robots. Challenges encountered during the study included data quality issues and the complexity of modeling industrial robot behavior. Future research should focus on improving the reliability of predictive models and developing new machine learning techniques for recognizing faults in industrial settings.

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